

Voting prediction using LLMs

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Abstract

This project explores the capacity of large language models (LLMs), in particular ChatGPT, in predicting voting behavior in Swiss popular votes. Using the “1231 VOTO Cumulative Dataset”, which contains both structured survey data and open-ended textual responses, three different models were developed and compared: a ChatGPT Text Model, which only uses open-ended text responses, a ChatGPT Numeric Model, which only uses numerical variables as an input and a Combined Model, which uses both textual and numerical input. As a benchmark, a traditional logistic regression model was created. Results show that while the ChatGPT Text Model (74.3% overall accuracy) and Logistic Regression Model (70.5% overall accuracy) perform moderately well, the ChatGPT Numeric Model (50.7% overall accuracy) struggles to correctly predict voting in favor or against a proposal. The Combined Model drastically outperforms the other models and reaches an overall accuracy of 83.2% in this project. A key innovation of this study is that the LLMs are not only used for predicting “yes” or “no” votes but also for generating text-based rationals behind each classification, which adds a valuable qualitative dimension. These results show the potential of LLMs in political research and could be inspirational for future work that uses hybrid modeling approaches.

1 1. Introduction

Voting behavior and especially predicting how people vote has long been a central topic in political science. Traditionally, predicting and analyzing voting behavior is carried out using regression-based models. These models use socio-demographic and political predictors to model voting behavior. Due to their quantitative nature they can indicate associations / correlations between variables but do not elucidate the underlying rationale behind an individuals decision.

In contrast to this, large language models such as ChatGPT made it possible to produce both predictions and explanations for these predictions. This offers a richer and much more interpretable perspective on decision-making when voting.

This paper compares four different approaches to predict individual vote choice in Swiss popular votes:

1. **ChatGPT Text Model:** Leverages open-ended text responses to predict whether a voter supported or opposed a proposal. In addition to providing a numeric classification, the model outputs a brief natural language explanation that clarifies the reasoning behind its prediction.
2. **ChatGPT Numeric Model:** Uses a carefully engineered prompt based solely on structured numeric indicators such as age, decision timing, political orientation, income, education, institutional trust, and media use. The model returns a numeric prediction along with a short textual explanation.
3. **ChatGPT Combined Model:** Integrates both numeric data and open-ended text inputs to capture a broader context, thus enabling a more nuanced prediction coupled with an explanation.
4. **Logistic Regression Model:** A classical quantitative approach that employs the same structured predictors as the ChatGPT Numeric and Combined models. This model serves

as a baseline; however, it does not incorporate any open text data or provide explanatory output.

In order to conduct these analyses, the “1231 VOTO Cumulative Dataset” was prepared, removing all of the irrelevant variables and cases. This micro-level dataset not only provides all the variables needed to conduct a traditional regression analysis but also provides multiple open text answers which can be used for the LLM-based analyses.

A key innovation of this work is the use of ChatGPTs ability to provide a text-based explanation next to the numeric prediction. This feature makes it possible to not only understand the outcome but also the rationale behind each prediction. The traditional regression model does not offer such qualitative insights.

In the following sections, data sources, variable selection, methodology, model implementation and performance are described before concluding the paper with a discussion of its implications.

2 2. Data and Methods

2.1 2.1 Data Source and Description

The data used in this project is from the “1231 VOTO Cumulative Dataset - Scrutin” (“VOTO Studies: Standardized Post-Vote Surveys, 2016-2020” n.d.). This dataset was chosen as it provides individual-level post-vote survey data from Swiss citizens collected after federal popular votes (ebd.). In contrast to other datasets, text answers are included in a non-recoded version which makes it interesting for my analyses (ebd.). I first wanted to choose the SELECTS dataset (“Swiss Election Study (Selects), Cumulative Dataset 1971-2019” n.d.) but unfortunately its text answers are all recoded.

The VOTO project is a collaboration between FORS, the Centre for Democracy Studies Aarau (ZDA) and other partners (“VOTO Studies: Standardized Post-Vote Surveys, 2016-2020” n.d.). It offers a rich basis for both traditional and large language model analysis due to its combination of survey data with open-ended text responses (ebd.).

My original plan, to predict voting and non-voting had to be abandoned due to the lack of responses / open-ended text answers of people that did not vote. The VOTO Dataset only includes open-ended text responses of people who voted. Therefore I changed my plan and decided to predict with different models if someone votes in favor or against a proposal.

Each respondent may have participated in up to five different popular votes. For each vote, the dataset contains:

- **Voting decision (vote_1):** This variable indicates if the respondent voted in favor of the proposal (1), against the proposal (2) or did not participate in the vote (3).
- **Proposal title (proposalx):** This variable describes the title of the corresponding proposal. Originally, it was intended to only use respondents referring to a specific vote, but as the number of open-text responses is too few, all of the proposals are taken into consideration.

- **Open-ended justification (e.g. reason1_acc1_txt):** In different open-ended text answers, respondents could explain their main and secondary reasons for supporting (reason_1_acc1_txt, reason_2_acc1_txt) or opposing (reason1_den1_txt, reason_2_acc1_txt) the proposal. These text answers were the central variables for my ChatGPT Text Model.

The dataset also contains several numeric variables, which I used for my baseline regression, the ChatGPT numeric model and the ChatGPT text and numeric model.

The variables used are:

- **birthyear:** Year of birth of the respondent, used to get the age of the respondent
- **dectime1:** Timing of the decision on what to vote. This nominal variable is coded as follows:
 - 1 = clear from the beginning,
 - 2 = during the campaign,
 - 3 = at the last moment,
 - 8 = don't know,
 - 9 = no answer.
- **lrsp:** Left-right political self-placement on a 0 (far left) to 10 (far right) scale.
- **income:** Grouped household income in groups from 1 (less than 2'000 CHF) to 15 (more than 15'000 CHF).
- **educ:** Educational level, coded from 10 to 60, with increasing levels of education.
- **trust_1:** Trust in Feder Council (0-10)
- **importance_1:** Importance of the vote (0-10)
- **mediause_3:** Use of TV debates (0-10)
- **mediause_1:** Use of newspaper articles (0-10)

This selection of variables makes the dataset ideal for testing traditional statistical methods such as the regression analysis used here aswell as models using large language models.

2.2 2.2 Data Preprocessing and Cleaning

The preprocessing and cleaning of the data is designed to align with the requirements of the four different approaches used in this project: Three large language model based approaches (only text, numeric and combined) and the traditional logistic regression model. The objective of the preprocessing was to reduce the dataset in order to work more efficiently, only keeping the variables and cases needed.

Only cases with complete information across all of the selected numeric variables were kept in the dataset. This is to make sure that all models can be evaluated in a consistent manner.

For the analyses using text variables, only respondents who provided at least one non-empty response in one of the selected text variables were retained. This ensures that all of the models operate on a meaningful and comparable input.

In order to make the output of the analyses more interpretable, two variables were created:

- **age:** Computed as 2020 - birthyear (in the dataset), to get the respondent's age at the time of the dataset's reference point
- **voted_flag:** Recoded version of vote_1 indicating whether someone voted in favor of the proposal (1) or against the proposal (2).

2.3 2.3 Modeling Approaches

Four different model approaches are used in this model in order to predict individual vote choices. Three approaches test the capabilities of large language models in predicting voting behavior. Each of the models works with a different type of input, which is explained in the following section. The fourth model, the traditional logistic regression model, serves as a benchmark.

2.3.1 2.3.1 ChatGPT Text Model

The ChatGPT text model only uses the open-ended text responses which were provided by respondents to justify their voting decision. The model receives input in form of written justifications for voting in favor or against a proposal and converts these into a numeric output. The output consists of the values 1 (predicting a vote in favor), 2 (predicting a vote against) and 99 (not enough information to make a prediction). Additionally the model also generates a text output explaining the factors influencing its classifying decision. This form of output adds a qualitative dimension which can not be achieved using traditional regression models.

Overview:

- Input: Open-ended text responses (e.g., reasons for/against voting)
- Output: Numeric classification (1, 2, or 99) plus a brief explanation

2.3.2 2.3.2 ChatGPT Numeric Model

This model is designed to work with structured data only. It uses all of the numeric variables shown before but no text variables. The model is prompt-engineered to interpret these quantitative measures, returning a numeric prediction with the values 1 (predicting a vote in favor), 2 (predicting a vote against) and 99 (not enough information to make a prediction). Again, the model also provides a brief explanation for its prediction. The goal of this approach is to showcase the potential of LLMs in the transformation of quantitative into both a prediction and a qualitative insight.

Overview:

- Input: Structured numeric indicators (e.g., Age, Decision Time, Political Left-Right, Income, Education, Trust, Importance, and Media Use)
- Output: Numeric prediction with natural language explanation

2.3.3 2.3.3 ChatGPT Combined Model

This model is a combination of the two first models. This model integrates both the qualitative open-ended text responses and the numeric indicators. By combining the richness of textual information together with quantitative measures, this model tries to capture a more holistic view of factors that determine voting behavior. Like the other two LLM based approaches the output is numeric with the values 1 (predicting a vote in favor), 2 (prediciting a vote against) and 99 (not enough information to make a prediction) aswell as textual in form of a brief explanation for its prediction.

Overview:

- Input: Integration of numeric indicators with available open-ended text responses for richer context
- Output: Numeric prediction with explanatory text

2.3.4 2.3.4 Logistic Regression Model

In order for it to serve as a reference model, this logistic regression model uses the same set of numeric variables as the ChatGPT Numeric Model and as the ChatGPT Combined Model. This statistical approach computes predicted probabilities and generates a classification based on the relationships between the independent variables and the dependent variable. As in the ChatGPT numeric Model, no input in form of text is provided. In this case, this would be impossible to do. Although this model does not provide any type of qualitative insight, logistic regression offers a well-established framework which enables a robust quantitative benchmark in order to evaluate the performance of the other models in this project.

- Input: The same numeric and categorical predictors used in the ChatGPT Numeric Model
- Output: Predicted probability and classification (without any open text explanation)

3 3. Analysis and Results

3.1 3.1 Model Evaluation Metrics

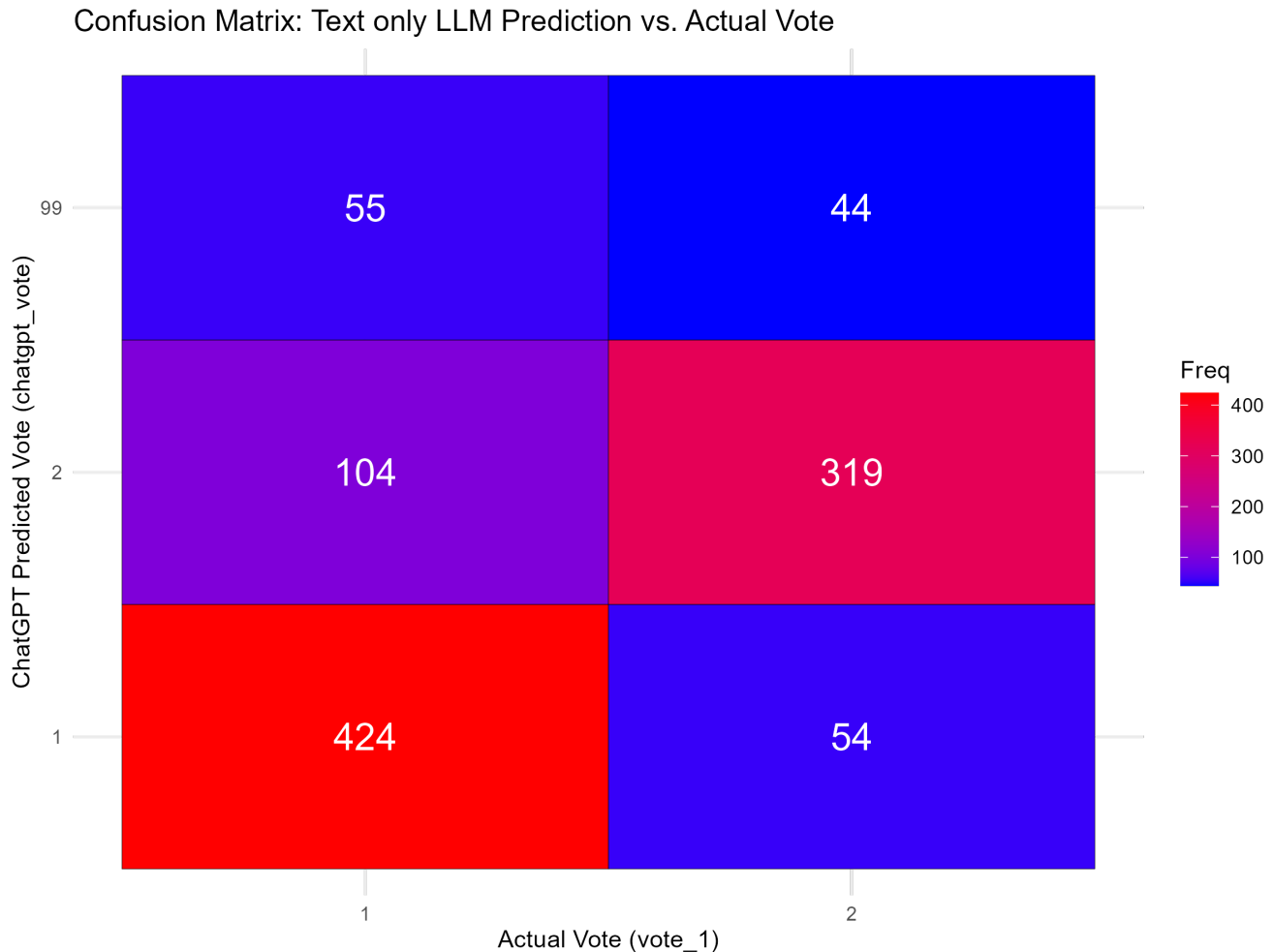
To evaluate the performance of the different models, I used a standardized evaluation script for all of the models. The actual vote (vote_1) of each respondent as well as the predicted vote from the model were first converted into factors in order to be able to construct a confusion matrix. This matrix compares the actual responses with the predictions and provides an overview over correct and incorrect classifications.

For better visibility, the confusion matrix was then visualized as a heatmap using ggplot2, which shows the frequencies of correct classifications / predictions alongside misclassifications. The main metric to evaluate the models is overall accuracy, which is computed by summing the correct predictions and putting them in relation with the total number of cases. Although overall accuracy is a great metric, it needs to be accompanied by false positive and false negative percentages as otherwise the classification performance of each model can not be properly evaluated.

These evaluation results are presented in a systematic order below, in order to highlight each models predictive strengths and possible weaknesses.

3.2 3.2 Results for Each Model

3.2.1 3.2.1 Results for the ChatGPT Text Model



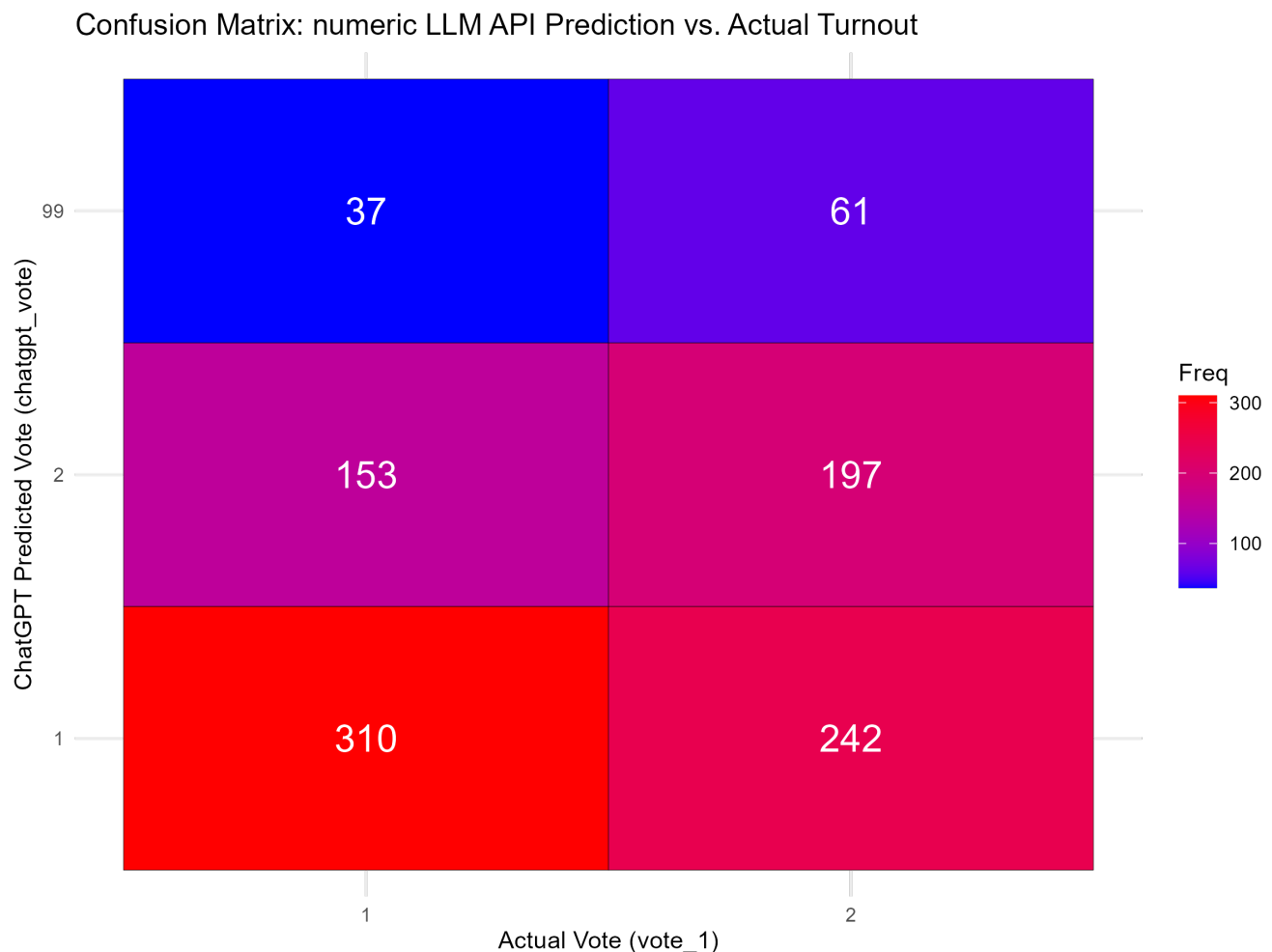
The ChatGPT Text model, which bases its predictions only on the open-ended text input, achieved an overall accuracy of 74.3%. The heatmap (figure above) shows that the model correctly classifies most of the respondents based on their text input. However, the heatmap also displays some misclassifications.

From the error metrics, class 1 (voting in favor) shows a false positive rate of 5.4% and a false negative rate of 15.9%. This indicates that while the model sometimes confuses genuine “yes” votes with “no” votes, it more frequently fails to recognize some actual “yes” votes. In contrast,

class 2 (voting against) exhibits a 10.4% false positive rate and a 9.8% false negative rate, which is a more balanced distribution of errors. Still, also here there is room for improvement.

These results suggest that textual inputs alone can capture valuable signals regarding respondents' motivations, but there is the danger of ambiguities in language. When individuals express nuanced or less explicit reasons, the model tends to misclassify these respondents. Nonetheless, this model provides an interpretative advantage by revealing the rationale behind each prediction which also means highlighting potential areas of confusion.

3.2.2 Results for the ChatGPT Numeric Model

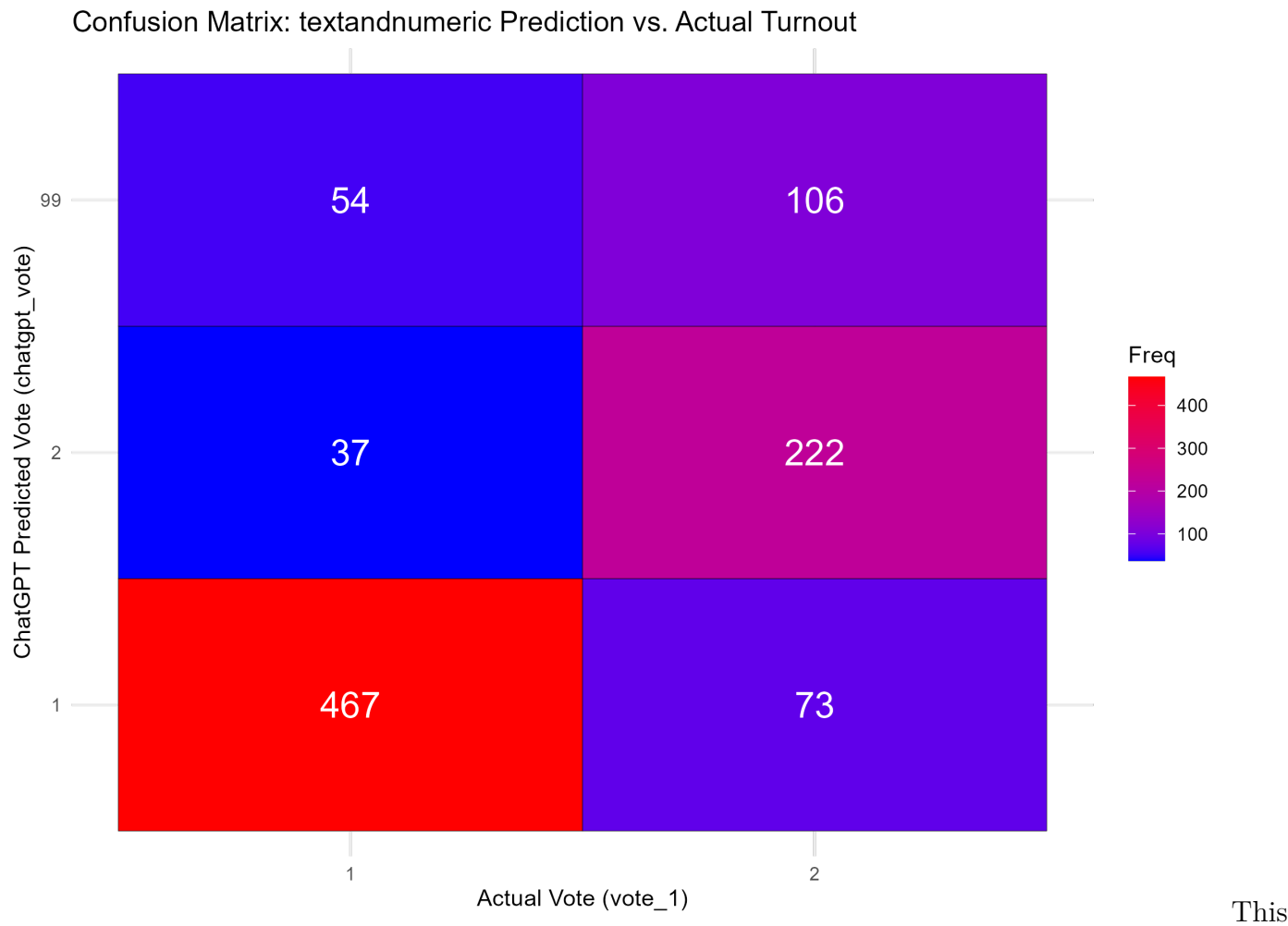


The ChatGPT Numeric Model, which only relies on numeric indicators such as age, political orientation and other demographic variables, achieved an overall accuracy of 50.7%. The heatmap

displays that the model struggles to classify many respondents' voting behavior, suggesting that the LLM cannot base its predictions solely on numerical input.

From the error metrics, class 1 (voting in favor) shows a false positive rate of 24.2% and a false negative rate of 19%. This implies that the model mislabels some of the “no” votes as “yes” votes but more often fails to correctly identify actual “yes” votes. These figures highlight the limitations of my model, which only provides numerical input and no further explanation regarding the topic of the proposals being voted on.

3.2.3 3.2.3 Results for the ChatGPT Combined Model



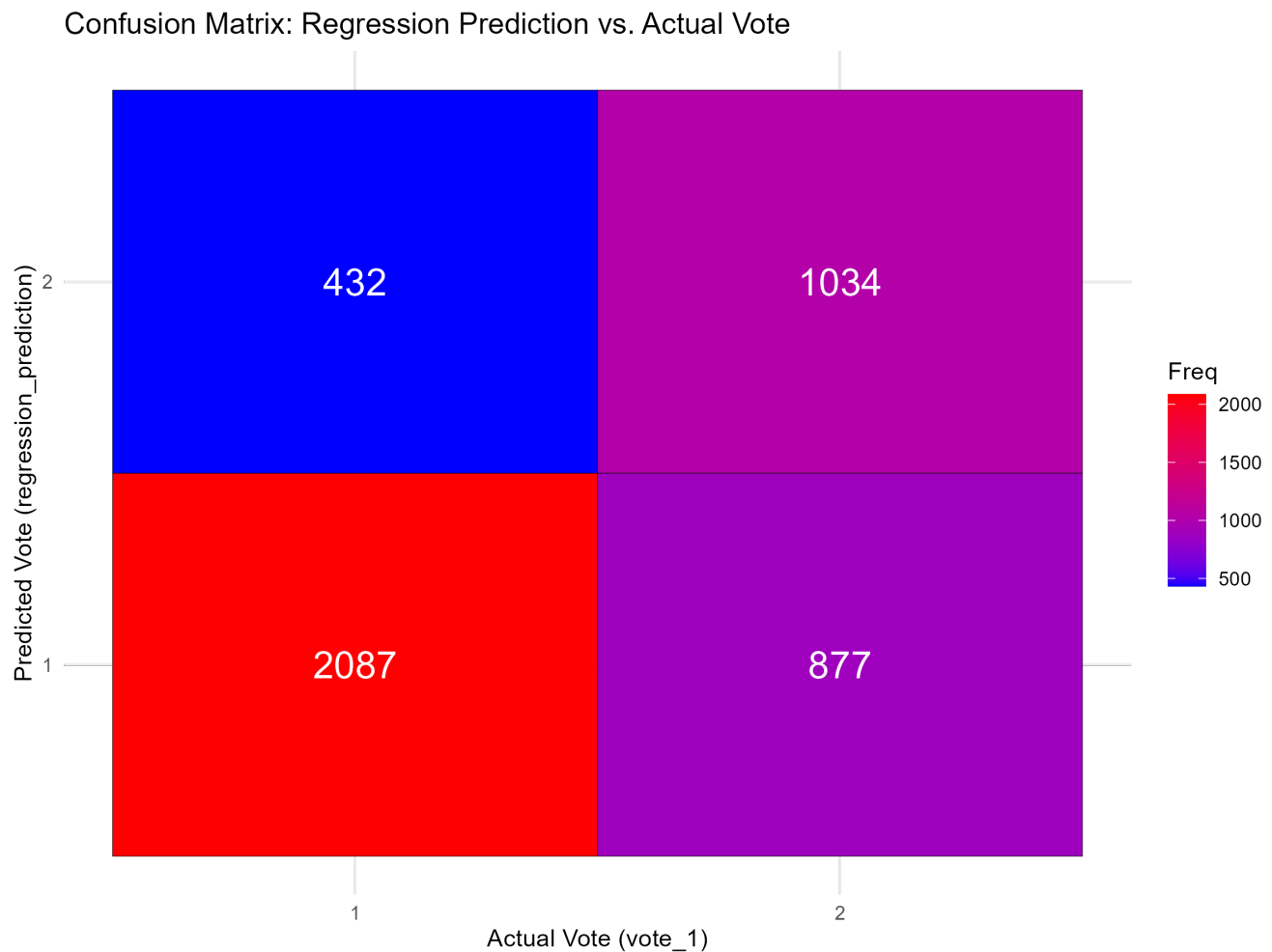
model, which combines both open-ended textual justifications aswell as numeric indicators achieved an overall accuracy of 83.2%. As shown in the heatmap above, correct classifications cluster along

the main diagonal of the confusion matrix, which highlights that the combination of qualitative and quantitative data enhances the model’s predictive performance.

Regarding the error metrics, class 1 (voting in favor) has a false positive rate of 3.2% and a false negative rate of 11.1%, while class 2 (voting against) shows a false positive rate of 6.4% and a false negative rate of 5.8%. These numbers suggest that the combined approach not only improves the overall accuracy but also reduces misclassification rates by mitigating the ambiguities which arise with the usage of one data type only.

Overall, these results indicate that combining numerical and textual input offers a more robust framework for capturing the complex nuances of voting behavior. The enhanced performance, as demonstrated by the concentrated clustering in the heatmap and the improved error metrics, underscores the importance of combining multiple data types as an input for prediction models.

3.2.4 Results for the Logistic Regression Model



The logistic model, which serves as a traditional reference for the other models, achieved an overall accuracy of 70.45%, using the same numeric predictors as the ChatGPT Numeric approach. As shown in the confusion matrix above, correct classifications line up more prominently along the main diagonal compared to the numeric LLM approach. Still, a considerable number of misclassifications exists.

From the error metrics, class 1 (voting in favor) exhibits a false positive rate of 19.8% and a false negative rate of 9.75%. Class 2, (voting against) has a false positive rate of around 9.75% and a false negative rate of 19.8%. This inversion is normal for binary classification as any error that misclassifies a “no” vote as a “yes” note or vice versa simultaneously counts as a false positive for one class and a false negative for the other. This did not occur in the other confusion matrices as

the value “99” was present.

Although this logistic regression outperforms the ChatGPT Numeric Model, these figures show that the logistic regression still confuses a notable share of “yes” votes for “no” votes and vice versa. This underlines the complexity of modeling voting decision making using only numeric data as input.

4 4. Discussion

In this project, four different approaches to predict individual vote choice using the Swiss VOTO dataset were used. The analysis shows the strengths and limitations of the used qualitative and quantitative models in capturing the complexities of voting behavior.

The ChatGPT Text Model, which only depended on open-ended textual responses provided by voters, reach an overall accuracy of 74.3%. This demonstrates that rich, unstructured text can capture some dimensions of voters' motivation for voting for or against proposals. As voters tend to articulate the reasons behind their decision in their own words the model can not only predict the way they vote but can also give a brief explanation for its prediction. However, this model struggles to sometimes classify certain respondents when their responses contain ambiguities in language.

In contrast, the ChatGPT Numeric Model, which only uses numeric indicators such as age, political orientation, income, level of education, trust in the federal council and the perceived importance of the vote, achieved an overall accuracy of 50.7%. The confusion matrix of this model shows that its predictions are rather scattered than grouped along the diagonal, which means that the model struggles to separate votes in favor from votes against. Moreover, the error metrics show that many actual "yes" votes were misclassified as "no" votes.

These findings indicate that neither only textual or only numerical data as an input is sufficient for accurately predicting voting behavior with LLMs in this project. While numerical data can capture measurable factors they overlook nuances that influence a vote. Similarly, textual responses, despite being rich in detail, can be ambiguous and inconsistent. This suggests that each type of data has certain limitations when used on its own.

The Combined Model, which uses both numerical data aswell as open-ended textual responses performed much better as it reached an overall accuracy of 83.2%. The confusion matrix for the combined model shows that the correct predictions are clustered along the main diagonal which

means that the model is strong at distinguishing between the two possible voting choices. by merging these two types of information, the model reduces misclassification errors and can better account for the factors that drive voting behavior. In addition to that, as for the ChatGPT Text Model, a brief explanation for the prediction is provided by the model which adds an interesting qualitative dimension to the numerical prediction.

Finally, the Logistic Regression Model which serves as a traditional statistical benchmark, achieved an overall accuracy of 70.45%. As the model only provides a binary prediction and no third “99” prediction, the confusion matrix shows a symmetric pattern of errors. Although the Logistic Regression Model performs better than the ChatGPT Numeric Model on its own, it fails to reach the effectiveness of the Combined Model. This does not come as a surprise, as this logistic regression, while robust and well-established, only provides quantitative estimates without any qualitative insights into a voters motivation.

5 5. Conclusion and Future Work

The findings of this project demonstrate that neither numeric nor textual data alone can fully predict voting behavior on an individual level. While numeric models can capture demographic factors, these models miss contextual nuances that often influence the way voters vote. Textual models are able to extract rich, qualitative insights from open-ended responses but their ambiguity sometimes leads to inconsistency in classifications. These observations suggested that moving on to creating a combined model may offer a better performance.

The Combined Model, which unifies numerical indicators and textual input, achieved the highest overall accuracy in this project, even outperforming the traditional Logistical regression. This dual approach not only improved the classification performance but also provides a more comprehensive understanding of voters decision making by combining measurable metrics with the context contained in the personal narratives provided.

Future work should try to refine these combined models. Advances in LLMs could improve the precision and consistency with which the textual input is analyzed. Newer ChatGPT models could make it possible to extract more accurate sentiment and context information from the text responses provided. Also improving the prompts which are used in this project or mentioning the topic of the proposal that is voted on, could add to improving the performance of this model. Using a different set of numerical indicators would also be a possibility to improve the models classification performance.

Ultimately, making these improvements in the combined model holds the potential to not only improve the models performance but also for a deeper understanding of the factors that drive individual voting choices, offering valuable insights for political analysis.

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